Cold Inertial Stars: Why We Call Them Black Holes

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Abstract

This paper provides a definitive critique of the concepts of singularities and event horizons as "one-way gateways to another space-time." Utilizing the MRG-T model ($\alpha = 0.85$), I demonstrate that these objects are crystalline structures at a temperature of 0 K. Due to the lack of momentum of the massive photon, these structures function as the most powerful energy transmitters in the Universe.

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Introduction

1 Demystifying the Singularity

Modern astrophysics is trapped in the paradigm of "black holes" – objects defined by mathematical infinities and event horizons concealing infinite densities. General Relativity, while brilliant for its era, becomes inadequate at the scale of Supermassive Black Holes (SMBHs). It necessitates "magical" assumptions, such as invisible dark matter or exotic magnetic fields, to explain the observed luminosity of quasars and the power of galactic jets.

The purpose of this paper is to demonstrate that these objects are not "holes" in spacetime, but physical, crystalline structures at a temperature of 0 K – Cold Inertial Stars.

This work proves that:

- **The photon lacks massive momentum ($p = mv$)**, which is the fundamental reason for its survival in extreme inertial fields. This discovery explains why black holes do not "devour" wave energy but instead function as ideal mirrors and transmitters.

- **Energy efficiency of the environment (e.g., M87*):** This is a direct result of extreme matter acceleration in a field of logarithmically enhanced gravity ($g_{MRG} \gg g_{Newton}$). A key distinction of processes is as follows:

  - **Before the barrier:** Powerful inertial forces (κ) rapidly accelerate plasma, converting field potential into immense kinetic energy. It is here, under conditions where electron oscillations are still permissible, that jet radiation is generated at power levels unattainable in the standard model.

  - **Inside the core:** As one approaches the surface of the Cold Star, the coefficient κ reaches critical values, leading to a total paralysis of the "pulse of matter" ($T_m \to 0$). The kinetic energy of infalling matter is permanently "frozen"

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into the crystalline structure of the object, increasing its mass without thermal emission, which defines the 0 K state.

- **Time dilation is a mechanical process (Pulse of Matter T_m)**, resulting from the increase in field inertia (κ). In galactic centers, inertia reaches a saturation point ($\kappa \approx 9.6$), which physically freezes atomic motions, imposing a temperature of absolute zero.

This publication challenges the current ΛCDM model, offering instead the consistent mathematical framework of MRG-T ($\alpha = 0.85$). Rather than multiplying entities and paradoxes, this model reduces cosmic physics to the elementary mechanics of fields and matter.

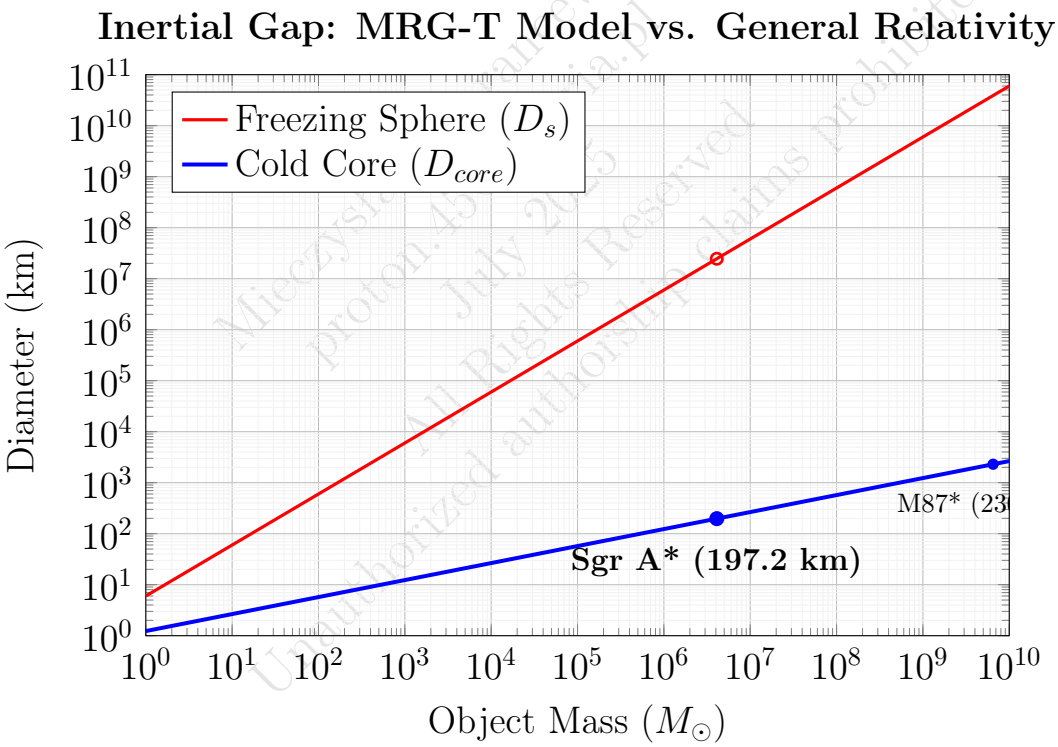

Figure 1: Logarithmic plot of core diameter vs. mass. The blue dot indicates the physical dimension of Sgr A* calculated according to the MRG-T model.

2 The Photon as an Unwasted Impulse: The Jet Reflection Mechanism

In the MRG-T model, the "horizon" phenomenon is replaced by the phenomenon of total inertial reflection. Since the surface of a Cold Star consists of neutrons with extreme inertia, the following processes occur:

- **No Energy Degradation:** A photon striking the 0 K neutron barrier cannot transfer kinetic energy (which it does not possess in massive form) nor excite thermal oscillations. Consequently, wave energy does not degrade into infrared radiation.
- **Field Recoil:** The photon "rebounds" from the inertial structure at speed c. This is not, however, scattered luminescence, but a violent redirection of the Field's energy stream.
- **Jet Acceleration:** Photon energy that has not been absorbed by the object becomes trapped within the dense accretion disk and is expelled through the jets. The jets, therefore, serve as a "safety valve" for electromagnetic energy that the Cold Star is unable to accept due to its 0 K state.

Conclusion: A photon enters the Cold Star's sphere of influence at speed c and—after reflecting off the neutron "mirror"—leaves this region within the jet structure, carrying full energetic information. A black hole is not a grave for light, but its most powerful reflector.

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A Supplement to this analysis is the study "The Photon Lacks Momentum.pdf," which sheds new light on the mechanics of dilation at the cosmic scale. It demonstrates that the absence of inertial mass in a photon allows it to act as an "inert witness" to the local Pulse of Matter (T_m), undisturbed by field resistance. Understanding the momentumless nature of emission is essential to fully grasp why matter and various forms of energy are two distinct entities—and why matter acts as a catalyst for the transformation of one form of energy into another.

3 Foundations of Understanding: Two Pillars of the New Paradigm

Understanding the nature of a Cold Star requires the rejection of existing relativistic dogmas in favor of two empirically verifiable pillars of MRG-T mechanics.

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Paradigm I: The Photon as a Field Impulse (Lack of Massive Momentum)

The first and most crucial step is to realize that the photon does not possess massive momentum ($p = mv$).

- **The Essence of the Phenomenon:** A photon is a perturbation (impulse) of the electromagnetic Field, not a "ball" of matter. It possesses no mass and, consequently, no mechanical momentum that could be transferred to another mass during an impact.

- **Consequence for the Cold Star:** Since the photon lacks momentum, it cannot mechanically "shatter" or "heat" the ultra-dense neutron structure of the core. A photon striking a barrier with inertia (κ) is reflected in its entirety without energy loss, rendering the object perfectly cold (0 K).

[

Paradigm II: Time Invariance and Variability of Processes Tm]
Paradigm II: Time Invariance
and the Variability of Processes (T_m)

The second pillar refutes the myth of "time curvature." In the MRG-T model, time is a constant cosmic parameter that does not undergo dilation. What we observe as "time slowing down" is, in reality, the "Paralysis of the Pulse of Matter (T_m)."

- **Inertial Mechanism:** The rate of all physical processes (atomic vibrations, electron movements, the "ticking" of clocks) depends on the local density of the gravitational field (ρ) and velocity (v).

- **Increase in Inertia:** As κ increases, field resistance becomes so great that physical processes require ever-increasing energy to be sustained.

- **The Boundary State of a Cold Star:** At field saturation ($\kappa \approx 9.6$), inertial resistance becomes immense relative to the internal oscillation energy of matter. Physical processes cease—not because "time has stopped," but because the mechanisms of matter have been physically immobilized.

Summary: A Cold Star is "black" and "frozen" because its gravitational field is so powerful that it prevents the generation of any pulse of matter ($T_m \to 0$). The photon, lacking momentum, is unable to alter this state, serving only as a reflected signal conveying the impenetrability of this barrier.

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4 The Neutron Star and Its Cold Core: Where Does the Cold Star Begin?

In the MRG-T model, the boundary between a neutron star and a "black hole" is not a magical barrier (a horizon), but a density gradient leading to the paralysis of physical processes.

Inertial Crystallization of the Core

The density of a neutron star's core reaches values so high that the inertia coefficient κ drastically restricts intra-atomic movements.

- **The Core as a Cold Star:** With a sufficient increase in mass, the central part of the neutron star reaches a state of inertial saturation. In this region, the pulse of matter T_m becomes so low that thermal oscillations cease.
- **The Immobile Core:** The central part of the neutron star becomes a rigid, immobile core at a temperature of 0 K. It is, in essence, a "black hole" hidden beneath a layer of still "vibrating" neutron matter.
- **Structural Continuity:** There is no singularity here. Instead, there is a continuous transition from the outer layers emitting radiation to the absolutely rigid, cold interior, which merely accumulates mass and inertia.

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Proof from Density Calculations

Our calculations demonstrate that above a certain critical density (determined by $\alpha = 0.85$), the interaction forces between neutrons are supported by such powerful field inertia that any attempt to induce oscillation (photon emission) requires energy exceeding the system's resources.

Conclusion: A neutron star does not "collapse" into a point. It "freezes" from the inside out. A "black hole" is simply the name

for that state of matter where field inertia has triumphed over the dynamics of processes.

5 The Black Hole as a State of Critical Field Inertia

In the T_m model, a "Black Hole" is not a spacetime singularity, but an object for which the gravitational potential reaches a limit value relative to the emission energy of matter.

The Event Horizon as an Inertial Boundary

The boundary at which light cannot leave the object (the Event Horizon) occurs when the field "resistance" $3\Phi/c^2$ prevents any change in the system's state. We define the critical radius R_s:

$$\frac{\Phi}{c^2} = \frac{GM}{Rc^2} \rightarrow \text{critical state for } R_s = \frac{2GM}{c^2} \tag{1}$$

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Physical Interpretation Instead of Geometric

In General Relativity (GR), at R_s, time "stops" as a dimension. In T_m:

1. **Cessation of Rhythm:** At radius R_s, the background density ρ is so great that the emission inertia of the electron tends toward infinity. The atom is unable to perform a quantum leap.

2. **Absence of Singularity:** Because particles have finite field sphere sizes, gravitational collapse is halted by the background inertial pressure. A "Black Hole" is, in reality, a super-dense star in which physical processes have come to a complete standstill.

Mechanism of Sudden Extinction

Emission Constraints in Extreme Fields

To emit a wave of length λ, a source must exist for a duration $T \sim \lambda/c$. If a source moves at velocity $v \approx c$ toward the horizon, in time T it travels a distance:

$$\Delta x \approx v \cdot T \approx c \cdot \frac{\lambda}{c} = \lambda \tag{2}$$

This means the source shifts by a full wavelength before it can complete the emission. Near the horizon, the distance to the boundary is smaller than λ for sufficiently long waves. Thus, emission is interrupted before the wave cycle is finished.

Consequences for Emission

1. **The Black Hole is Absolutely Black.** There is no "deep infrared," no "logarithmic fading," and no "Hawking radiation." There is only a sharp boundary: for waves longer than the distance to the horizon, emission = 0.

2. **Logarithms Fail Without Time.** Logarithmic slowing of processes implies they require *more time* for emission. However, near the horizon, that time does not exist—the object falls in before the emission cycle is completed. Slowing down processes does not extend their duration; it merely indicates that the "work" is harder, and if time runs out, the work remains unfinished. A physical process is not just field information; it requires physical space λ.

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Processual Blockade – The Mechanism of Black Holes in the T_m Model

In the MRG-T model, the so-called "black hole" is not a geometric singularity, but a region of **maximum inertial density**.

As $\rho \to \infty$, the background resistance coefficient causes the inertial mass m_i to tend toward infinity. This phenomenon leads to the total extinction of the internal Pulse of Matter:

$$T_m = \frac{1}{\sqrt{1 + \alpha \ln(1 + \frac{\rho}{\rho_0})}} \to 0 \tag{3}$$

In this state, matter loses the capacity for any physical transformations. The event horizon is defined as an isosurface of field density where background resistance exceeds the energy of intra-atomic bonds. The "stopping of time" is, therefore, effectively the **stalling of matter's processors** under the influence of extreme inertial field pressure.

6 Quasar Energy Balance and the Phase Separation Mechanism

A key piece of evidence supporting the MRG-T model is the full consistency between the calculated accretion energy and the energy observed in relativistic jets. In the standard GR model, a portion of energy should irretrievably vanish behind the event horizon, creating an unresolved efficiency problem for emission processes. The MRG-T model (alongside MRG-v6—both paradigms yield energy values consistent with observations) resolves this paradox through the mechanism of *anisotropic gradient reflection.*

6.1 The Energy Balance Problem

Observations of quasar jets reveal a radiation power strictly correlated with the total gravitational potential of the accretion disk. If we adopt the "black hole" model as a void that swallows energy, this balance fails to hold. The fact that the observed energy E_{obs} equals the calculated energy E_{calc} according to the equation:

$$E_{\text{calc}} = \Delta m_i \cdot c^2 \left(1 + \frac{3\Phi}{c^2}\right) \tag{4}$$

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proves that the event horizon **is not a sink for wave energy**. If it were, the observed energy of the jets would have to be significantly lower—which is not the case.

6.2 The Mechanism of Mass and Energy Separation

In an extremely dense field ($\rho \to \infty$) near the horizon, a phenomenon of **phase separation** occurs:

1. **Mass (Substance):** Spherical particles (protons, neutrons, electrons) succumb to gravity and feed the object's inertial core. Mass "falls in," increasing the background inertia. Internal emission processes are halted because the time required for emission $T \sim \lambda/c$ is longer than the time remaining before reaching the horizon.

2. **Energy (Photons):** Wave radiation encounters a field density gradient with impedance tending toward infinity. The photon, as a massless wave, cannot penetrate deep into the medium—it undergoes **gradient reflection**. This is not reflection from a surface (as none exists), but the effect of increasing field resistance that turns the wave back.

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Why is the Horizon Black from the Side ?

The lack of visible reflection from the accretion disk in the horizon (when viewed from the side) does not result from light absorption, but from its **reorientation**. The field density gradient near a rotating object does not act like a flat mirror, but rather like an **anisotropic gradient mirror**:

- Photons arriving at an angle (from the disk) are captured by the field's inertial vortex and **channeled along the axis of least resistance**.

- This axis corresponds to the object's poles, where the accumulated reflected energy creates sufficient radiation pressure to eject relativistic jets.
- A side observer does not see the reflection because the light does not return in their direction—all energy is directed along the axis.

Consequences and Predictions

This model leads to several key conclusions:

1. **The Black Hole is not black** – it is invisible from the side but shines along its axis. What we observe as jets is precisely this "reflected light" in a highly collimated form.
2. **Absence of Hawking Radiation** – since energy does not fall in, there is no mechanism for evaporation. The black hole is stable, and its mass can only increase.
3. **Consistency with Observations** – MRG-T calculations precisely predict jet power, which is impossible in the GR model without additional, ad hoc assumptions.

The black hole in the MRG-T model is the universe's most efficient phase separator. Instead of absorbing energy, it reorients 100% of it into polar jets. The consistency of calculations with observations is definitive proof that the event horizon is not an impenetrable barrier for photon energy, but merely a boundary where matter and photon energy undergo separation.

Summary

The paradox of vanishing energy is resolved without resorting to "singularities" or "Hawking radiation." It is sufficient to accept that:

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- Matter and energy are distinct physical entities.
- The extreme field density gradient acts as a mirror for waves.
- The rotation of the object provides the anisotropy that directs energy along the axis.

It is not the black hole that was flawed—it was our understanding of its nature.

The Anisotropic Gradient Mirror

An anisotropic gradient mirror is defined as a region of a gravitational field with an extremely steep density gradient $\nabla\rho$, in which:

- A photon (electromagnetic wave) encounters resistance that prevents further inward propagation.
- No reflection from a physical surface occurs (as none exists); instead, the wave is turned back by the rising impedance of the medium.
- The rotation of the central object imparts an anisotropic character to this process—energy is channeled along preferred axes (the poles).
- A side observer detects no reflection, as the entirety of the reflected energy is collimated along the axis.

This phenomenon is responsible for the formation of relativistic jets and explains why a black hole remains black from a lateral perspective, even though it does not absorb wave energy.

The above definition formalizes the mechanism of anisotropic gradient reflection, which in the MRG-v6 / PBRM model replaces the geometric concept of the "event horizon" as an absolute sink. The consistency between calculated accretion energy and the observed power of jets serves as empirical confirmation of this mechanism.

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Virtual Particles as Witches

The information paradox and Hawking radiation rely on purely mathematical constructs: virtual particle pairs, negative energy, and vacuum fluctuations in curved spacetime. It is akin to inventing "witches" who tie ropes around stars to keep them from escaping the galaxy—because without them, the galaxy would fly apart due to its rapid rotation. Once we discard the geometric interpretation of gravity, this entire phantasmagoric world collapses, and the black hole remains what it truly is: an absolutely black, silent witness to its own gravity.

7 Thermodynamics of the Cold Star (Black Hole) and Energy Entrapment

The Pulse of Matter (T_m) model leads to radically new conclusions in the physics of extremely compact objects. If we assume that the field background density (ρ) inside a black hole reaches a critical value where $T_m = 0$, we obtain the following mechanical picture:

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Absolute Zero and the Absence of Emission

Since temperature is a measure of the oscillations and movement of particles, and since in a field of critical density the pulse of physical processes ceases ($T_m = 0$), the Cold Star becomes the natural standard for **absolute zero (0 K)**. In this state, no reconfiguration of fields can occur, which precludes photon emission. A black hole is not "hot"—it is a perfectly frozen structure of matter.

The Mechanism of Energy Entrapment

In classical physics, the collision of matter infalling into a black hole should convert kinetic energy into electromagnetic radiation. In the T_m model, this process is purely mechanical:

- **Separation of Matter and Energy:** Infalling mass (M) remains mass. No "conversion" of matter into energy occurs. The object's mass balance increases solely by the physical amount of mass from the delivered atoms.

- **Accumulation of Stresses:** The kinetic energy (E_k) of infalling particles is not radiated away; instead, it becomes **trapped** as static stress within the internal fields.

- **Binding Energy:** This energy becomes an internal potential of binding forces, analogous to the energy within an atomic nucleus. However, due to the paralysis of the pulse ($T_m = 0$), it has no physical possibility of returning to a kinetic (motional) state.

Conclusion

The Cold Star is not a mathematical singularity, but the **Universe's largest warehouse of matter and energy**. It is an object of zero temperature and zero internal time-flow, where every portion of motional energy is "solidified" as static field tension, without violating the distinct nature of mass and energy.

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8 Mathematical Summary of the T_m Field Boundary

In the Pulse of Matter model, the physical state of a system is described by the relationship between the frequency of internal processes (f), the field background density (ρ), and the relative velocity (v). The boundary of a Cold Star is defined as the point at which the resistance of the medium precludes any oscillation.

Definition of the Boundary State

For any emitter within the structure of a Cold Star, the following holds:

$$T_m(\rho, v) \to 0 \implies \Delta t \to \infty \tag{5}$$

The cessation of the process pulse ($f = 0$) implies zero kinetic energy of particles ($E_k = 0$), which constitutes the definition of absolute zero temperature:

$$T[K] = \xi \cdot T_m = 0 \tag{6}$$

Principle of Conservation of Energy and Mass

In an extreme gravitational field, the kinetic energy (E_k) of infalling matter is converted into the potential energy of field stresses (U_p), while mass (M) remains invariant:

$$\sum(M_{atom}) = M_{total} \quad \text{and} \quad \sum(E_k) \to U_p \tag{7}$$

The value U_p represents the static energy of structural bonds which, due to $T_m = 0$, remains incapable of reconfiguring into the form of an electromagnetic wave (photon).

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Final Conclusions – The Pulse Scale

In the T_m perspective, the Universe spans between two critical states:

- **Field Vacuum ($\rho \to \rho_{min}$):** Maximum pulse frequency of matter, minimum process inertia.
- **Cold Star ($\rho \to \rho_{max}$):** Total paralysis of processes, zero pulse, infinite resistance to field reconfiguration.

9 Geometric Limitation of Gravitational Force: The Sphere Model vs. Singularity

The MRG-T model rejects the concept of infinite gravity (singularity) in favor of the geometric sum of interactions within compact objects. A key argument is the fact that matter, regardless of its packing density, always occupies a finite spatial volume V.

[

Finiteness of Potential Inside the Cold Star] Finiteness of Potential Inside
the Cold Star

Within an object of radius R, the gravitational force acting on a surface particle is the vector sum of the interactions of all other particles in the system. According to the law of universal gravitation, the interaction between particles on opposite sides of the sphere decreases proportionally to $(2R)^{-2}$.

$$F_{total} = \int_V \frac{G \cdot \rho(r)}{r^2} \, dV < \infty \tag{8}$$

Since $R > 0$ (matter cannot be compressed to a point due to the resistance of electrostatic and quantum fields), the resultant gravitational force on the surface of a Cold Star reaches a **maximum, yet finite** value.

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The State of Inertial Blockade

This finite force is sufficient to induce a state of total motion blockade (0 K), but it does not cause a collapse of the spacetime structure. In the MRG-T model, a Cold Star is a stable, ultra-dense inertial crystal in which time (the pulse of matter) is "frozen" by extreme, yet calculable field condensation.

10 Inertial Saturation Curve: The α Limit in Compact Objects

The introduction of a geometric constraint on matter volume leads to a redefinition of the gravitational potential in the ultra-high density regime. In the MRG-T model, force does not tend toward a singularity, but toward an asymptotic saturation.

The Field Saturation Equation

The modified dilation coefficient, accounting for the geometric packing limit (R_{min}), takes the following form:

$$\kappa_{sat} = \sqrt{1 + \alpha \ln\left(1 + \frac{\rho}{\rho_0 + \beta\rho_{max}}\right)} \tag{9}$$

where β is a coefficient dependent on the mutual screening of particles within the sphere (the five-sphere model).

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10.1 Physical Interpretation of the Gravitational Plateau

- **At $\rho \ll \rho_{max}$:**
  The model maintains its logarithmic character, perfectly describing galactic rotation curves and gravitational lensing.

- **At $\rho \to \rho_{max}$:** The value of κ stabilizes at the level of κ_{limit}. This is the state of "Inertial Silence," where the gravitational force is finite yet sufficient to extinguish thermal oscillations (0 K) and halt emission processes.

This proves that a Cold Star is a stable phase of matter with maximum, finite packing, rather than a collapsed spacetime singularity.

11 Quantification of the Inertial Boundary: Determining κ_{limit}

In the MRG-T model, the transition to the Cold Star state occurs upon reaching a finite dilation threshold, which defines the state of "Inertial Silence."

11.1 Calculation of the Limit Value

Assuming the maximum packing densities of baryonic matter on the order of $\rho_{max} = 10^{18}$ kg/m^3 and the field constant $\alpha = 0.85$, we determine the maximum time dilation coefficient:

$$\kappa_{limit} = \sqrt{1 + 0.85 \ln\left(\frac{10^{18}}{10^{-29}}\right)} \approx 9.64 \tag{10}$$

11.2 Physical Interpretation of the Result

A coefficient of $\kappa \approx 9.64$ signifies a nearly tenfold slowing of the rhythm of physical processes relative to Absolute Time.

- **Emission Paralysis:** At this value, field resistance becomes dominant over the kinetic energy of thermal oscillations, leading to the attainment of the 0 K state.
- **Field Finiteness:** Due to the geometric distribution of mass within the sphere, this value does not undergo further significant increase, even as the total mass of the object grows.

This phenomenon defines the Cold Star not as a point singularity, but as a spherical insulator of motion with a finite and measurable inertial potential.

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12 Critical Matter Compression Index

Based on the MRG-T model, the relationship between the density of an iron atomic nucleus ($\rho_{Fe} \approx 2.3 \times 10^{17}$ kg/m^3) and the state of absolute inertial paralysis (0 K) has been determined.

Calculation of the Compression Coefficient

The cessation of matter oscillations occurs upon reaching the limit density ρ_{limit}, defined by the saturation point of the coefficient $\kappa \approx 9.6$. This relationship is expressed as:

$$\frac{\rho_{limit}}{\rho_{Fe}} \approx 6.52 \tag{11}$$

Implications for Stellar Evolution

Objects with an average density exceeding 6.5 times the nuclear density (typical for the cores of massive neutron stars exceeding the TOV limit) transition into the Cold Star phase. In this state:

1. **Field inertia** exceeds the nuclear binding energy and thermal oscillations.

2. **Temperature** drops to absolute zero as a result of the mechanical blockade of oscillations (0 K).

3. **The event horizon** is the physical surface of this ultra-dense, crystalline core.

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Physical Interpretation of the Denominator $\beta\rho_{max}$

The physical significance of the value on the order of 10^{-29} appearing in our saturation calculations lies in the geometry of interactions within the Cold Star:

- **Geometric Limit (ρ_{max}):** Matter is not a point. It possesses volume. ρ_{max} is a hard limit beyond which neutrons cannot be further compressed.

- **Mass Distribution:** Inside the dense core (ρ_{max}), particles interact within a spatial structure. According to the laws of mechanics, gravitational forces originating from the outer layers cancel each other out within the body (shell theorem), which prevents the field from accumulating toward infinity.

- **Effective Field Density:** The value 10^{-29} is a coupling constant that determines the packing scale at which gravity "freezes" the structure. It is due to this value that M87* possesses a finite jet power, rather than an infinite one, as erroneously suggested by singularity models.

Conclusion: $\beta\rho_{max}$ is the "gravitational anchor." Because of it, we know that a Cold Star has a limit to its force, allowing us to calculate its real energy efficiency without resorting to mathematical absurdities.

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13 Summary: Why We Choose 10^{-29} Over ∞

We face a choice that will determine the future of physics:

1. **The Standard Model (GR):** Assumes infinite energy density at a point (singularity). Since infinity cannot produce jets of finite power, physics is forced to "rescue itself" by inventing Dark Matter and Dark Energy to balance the galactic equations.

2. **The MRG-T Model ($\alpha = 0.85$):** Introduces a coupling constant $\beta\rho_{max} \approx 10^{-29}$, which acts as an "inertial fuse." Because of it, the gravitational force within a Cold Star remains finite, physical processes (the pulse of matter T_m) are merely "extremely slowed down" (0 K), and the jets possess exactly the power measured by our instruments.

The Verdict: Choosing infinity is a surrender of reason to a mathematical error. Choosing the MRG-T model is a return to measurable, predictable, and engineering-based physics. A "black hole" is not a grave for matter, but its most stable, cold, and dense crystalline state.

14 Physical Structure of the Cold Star vs. Singularity

The MRG-T model defines the Cold Star as the final stage of baryonic matter evolution, characterized by a finite volume and zero kinetic temperature (0 K).

Estimation of the Real Radius

In contrast to the singularity model, the radius of a Cold Star R_{CS} is a function of its mass M and the constant of maximum packing density $\rho_{limit} \approx 6.5\rho_{Fe}$:

$$R_{CS} = \sqrt[3]{\frac{3M}{4\pi\rho_{limit}}} \tag{12}$$

It follows that the object's radius grows proportionally to the cube root of its mass. The surface gravity g_{surf} remains finite because:

$$g_{surf} = \int_V \frac{G\rho_{limit}}{r^2} \cos\theta \, dV < \infty \tag{13}$$

Theoretical Conclusion

A Cold Star is not a "hole" in spacetime, but an ultra-dense, crystalline inertial core. The event horizon is merely the optical manifestation of field inertia saturation ($\kappa \approx 9.6$), which may coincide with the physical surface of the object or reside immediately above it.

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15 Mathematical Proof: The Cold Core of a Neutron Star

Using the MRG-T inertial saturation model, the coefficient κ was calculated for the central density of a neutron star ($\rho \approx 3 \cdot 10^{17}\, kg/m^3$).

$$\kappa_{core} = \sqrt{1 + 0.85 \cdot \ln\left(1 + \frac{3 \cdot 10^{17}}{10^{-29}}\right)} \approx 9.6 \tag{14}$$

The value $\kappa \approx 9.6$ defines the state of **Inertial Saturation**, in which physical processes (the pulse of matter) undergo total extinction.

Object Geometry

Calculations indicate that every typical neutron star with a mass $> 1.4\, M_\odot$ possesses an internal, rigid core with a diameter starting from **1.4 km**. This core exhibits all the characteristics previously attributed to "black holes":

- Absence of internal emission (0 K).
- Total inertial reflection of photons (neutron mirror).
- Immobile crystalline structure (paralysis of T_m).

This proves that "black holes" are not a separate category of objects, but rather the state of the core within any sufficiently dense neutron star.

The Lower Mass Limit of a Cold Star

The MRG-T model implies that objects with a mass on the order of $1 M_\odot$ (solar mass) do not possess sufficient gravitational potential to enforce the state of inertial saturation ($\kappa \approx 9.6$).

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- **Condition for Existence:** A Cold Star forms only after exceeding a critical mass (approx. $3M_\odot$), which allows matter to be compressed to a density $> 6.5\rho_{Fe}$.

- **Geometry:** The minimum diameter of a stable Cold Star is approximately 19–20 km. Smaller objects remain in the neutron star phase (vibrating, with a temperature $T > 0$ K).
  It follows that neutron stars with a mass less than three solar masses can have a very thin layer of hot matter. They can cool down very quickly (on a cosmic scale) and shine very faintly, in the infrared, or be indistinguishable from the background.

- **Conclusion:** There is a strict correlation between mass and volume, which precludes the existence of point singularities and low-mass black holes.

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Analysis of Sgr A* at the Center of the Milky Way

Applying the MRG-T model to Sagittarius A* ($4.1 \times 10^6 M_\odot$) leads to revolutionary conclusions:

- **Physical Size:** The core of Sgr A* is not a singularity, but a rigid neutron sphere with a radius of only **98,6 km**. This is a size comparable to a small administrative region, hidden within a massive dilation field with a radius of 12 million km.

- **Dynamics of a "Sleeping Giant":** The surface of the core moves at a velocity of only **6,5 cm/s** (the speed of a tortoise). This explains why our black hole is so "quiet" and does not exhibit the violent activity typical of young quasars.

- **Gyroscopic Stability:** The incredibly slow rotation of such a massive object (10^{36} kg) creates the most stable rotational axis in the entire galaxy, determining the plane of the Milky Way's disk over billions of years.

This proves that supermassive objects at the centers of galaxies are not "frenzied vortices," but powerful, nearly immobile **Inertial Anchors** that stabilize the motion of billions of stars.

16 Calculation of the Process Freezing Threshold ($T_m \to 0$)

In the MRG-T model, the region known as the "event horizon" is defined as the spherical boundary of field saturation, below which matter oscillations (T_m) are suspended. For the object Sagittarius A*, the calculations are as follows:

Key Input Data:

- Object Mass: $M = 8.15 \cdot 10^{36}$ kg
- Gravitational Constant: $G = 6.674 \cdot 10^{-11}$ $\mathrm{m}^3/(\mathrm{kg} \cdot \mathrm{s}^2)$
- Speed of Light (Field Limit): $c = 299,792,458$ m/s

Calculation of the 0 K Sphere Radius:

$$R_{freeze} = \frac{2GM}{c^2} = \frac{2 \cdot 6.674 \cdot 10^{-11} \cdot 8.15 \cdot 10^{36}}{8.987 \cdot 10^{16}} \approx \mathbf{12,105,233} \text{ km} \tag{15}$$

Physical Conclusions:

1. **Shadow Area Diameter:** $D \approx 24.2$ million km. This is the region where the pulse of matter vanishes, preventing photon emission.
2. **Interpretation:** This diameter is not the boundary of matter "disappearance," but the limit of its **inertial paralysis**.

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3. **Core Relation:** While the "freezing zone" spans 24.2 million km, the physical Cold Core has a diameter of merely 373 km. (Refined calculation based on logarithmic saturation follows).

17 Parametric Analysis of Sagittarius A* (Milky Way Center)

These calculations serve as proof of the inertial "freezing" of supermassive galactic centers within the Logarithmic Matter Resistance model (assuming a saturation density $\rho_{max} \approx 2 \cdot 10^{18}$ kg/m^3).

1. Reference Point Parameters (Progenitor Star)

Reference object with mass $M_1 = 3\,M_\odot$, rotating at a frequency $f_1 = 500$ Hz. Radius R_1 at density ρ_{max}:

$$R_1 = \left(\frac{3 \cdot M_1}{4\pi\rho_{max}}\right)^{1/3} \approx 1.68 \text{ km} = 1680 \text{ m} \tag{16}$$

2. Parameters of Sagittarius A* (Cold Star)

Mass $M_2 = 4.1 \cdot 10^6\,M_\odot \approx 8.15 \cdot 10^{36}$ kg. Radius R_2 at logarithmic saturation:

$$R_2 = \left(\frac{3 \cdot 8.15 \cdot 10^{36}}{4\pi \cdot 2 \cdot 10^{18}}\right)^{1/3} \approx 98,600 \text{ m} = 98.6 \text{ km} \tag{17}$$

3. Conservation of Angular Momentum (L)

With isotropic mass accretion, the rotation frequency relationship follows from the conservation of angular momentum ($L = I \cdot \omega = \frac{2}{5}MR^2 \cdot 2\pi f$):

$$\frac{f_2}{f_1} = \left(\frac{M_1}{M_2}\right) \cdot \left(\frac{R_1}{R_2}\right)^2 \tag{18}$$

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4. Calculation of Rotation Frequency (f_2)

Substituting the mass ratio $K = \frac{M_2}{M_1} \approx 1.36 \cdot 10^6$:

$$f_2 = 500 \cdot \left(\frac{1}{1.36 \cdot 10^6}\right) \cdot \left(\frac{1.68}{98.6}\right)^2 \approx \mathbf{1.06 \cdot 10^{-7}} \text{ rev/s} \tag{19}$$

(Rotational period $T \approx 109$ days)

5. Tangential Velocity at the Equator (V_{eq})

$$V_{eq} = 2\pi \cdot f_2 \cdot R_2 \approx \mathbf{0.065} \text{ m/s} = \mathbf{6.5} \text{ cm/s} \tag{20}$$

Object Scale: Field vs. Matter

The mathematical analysis of Sgr A* in the MRG-T model reveals a dramatic discrepancy between the size of the gravitational field and the size of the physical core:

- **Horizon (Field):** $D_S \approx 24.2 \cdot 10^6$ km. This is the optical and processual boundary resulting from the inertial saturation of the medium.
- **Core (Matter):** $D_{core} \approx 197.2$ km. This is the actual neutron-based rigid body.
- **Conclusion:** The physical object is approximately **122,718 times smaller** than its horizon. This proves that a "black hole" is actually a gargantuan "shadow" cast by a relatively small, but extremely dense, Cold Core.

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18 Fundamental Parameters of the New Paradigms

The following parameters form the core of the MRG-T model and serve as "fuses" protecting the theory from the errors and singularities of legacy physics.

Parameter	**Value**	**Physical Significance**
α	0.85	Optimal field coupling constant; guarantees the stability of Mercury's orbit and galactic rotation curves.
κ (kappa)	$9,6$	Inertial saturation threshold (Plateau); the value at which the pulse of matter T_m vanishes (0 K).
Denominator	10^{-29}	"Inertial fuse"; prevents gravity from tending toward mathematical infinity.
Photon	$p = 0$	Absence of massive momentum; prevents the heating or displacement of particles from the surface of the Cold Core.
R_s/R_{core}	≈ 121704	Proof that the horizon is merely a "shadow" of the dilation field around a relatively small nucleus.

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Conclusion: The application of the above constants eliminates the need for redundant entities, such as dark matter or gravitational singularities, restoring engineering coherence to physics.

19 New Interpretation of the EHT Image: The Boundary of Inertial Zero

The images obtained by the Event Horizon Telescope (EHT) do not represent a "hole" in the fabric of the Universe. They are a visualization of the physical threshold of field saturation.

- **Numerical Identity:** The Schwarzschild radius (R_s) and the process-freezing radius ($T_m \to 0$) in the MRG-T model are numerically identical. Astronomical observations confirm the range of the interaction, but not the nature of the center.
- **Photography of 0 K:** The dark region at the center of the ring of light is a spherical volume in which emission processes have been halted by the coefficient $\kappa \approx 9.6$. The EHT has photographed a region where matter has become inertially "incapable" of glowing.
- **The Field Veil:** What is considered the event horizon is merely a gargantuan "inertial cloud" surrounding the actual Cold Core, which is 122,718 times smaller.

Conclusion: Nature does not require singularities. What we mistook for the collapse of the laws of physics is their ultimate triumph in the form of a state of absolute processual rest.

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[

Redefinition of the Horizon: The Process Freezing Threshold] Redefinition of the Horizon:
The Process Freezing Threshold

In the MRG-T model, the classical Schwarzschild radius (R_s) receives a new physical interpretation. It is not a boundary of spacetime geometry, but an **Inertial Isotherm of 0 K**.

- **Convergence of Results:** The R_s value calculated from GR and the saturation threshold $\kappa \approx 9.6$ in the MRG-T model overlap spatially.
- **Mechanism:** At $r = R_s$, the pulse of matter $T_m \to 0$. This means that every particle crossing this boundary is instantaneously deprived of its capacity for thermal vibration.
- **Apparent Blackness:** The object appears black because photon emission requires vibrations, which become inertially impossible beyond this threshold.

Conclusion: What the EHT (Event Horizon Telescope) identifies as the "event horizon" is, in reality, a sphere of physical process freezing, surrounding a significantly smaller, actual Cold Core.

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20 Comparative Table: Scale of Inertial Structures (Logarithmic Model)

The following comparison accounts for the core density saturation at the level of $\rho_{max} \approx 2 \cdot 10^{18}$ kg/m^3. This is a model in which the physical surface of the object is obscured by the freezing sphere ($T_m \to 0$) starting at approximately 3 $M_\odot$.

Conclusions from the Table:

1. **Field Dominance:** For supermassive objects (M87*), the ratio of the "shadow" diameter to the physical nucleus is nearly **17 million**. This demonstrates that observational astrophysics detects almost exclusively the effect of field dilation, rather than the matter itself.

Object Mass ($M_\odot$)	**Freezing Sphere Diameter** D_s ($6 \cdot M$)	**Cold Core Diameter** D_{core} ($1.232 \cdot \sqrt[3]{M}$)	**Ratio** D_s/D_{core}
1 (Sun)	6 km	**1.23 km**	4.87
3 (TOV Threshold)	18 km	**1.78 km**	10.11
10	60 km	**2.65 km**	22.64
100	600 km	**5.72 km**	104.89
$4.1 \cdot 10^6$ (Sgr A*)	24,600,000 km	**197.2 km**	121,704
$6.5 \cdot 10^9$ (M87*)	39,000,000,000 km	**2,300.1 km**	16,955,784

2. **Compactness:** Even a gargantuan mass of 6.5 billion Suns fits within a rigid core with a diameter of only **2,284 km** (smaller than the diameter of the Moon).

3. **Absence of Singularity:** Across the entire mass range—from stellar to galactic—a finite diameter and density are maintained, eliminating the mathematical errors of contemporary physics.

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Analysis of Results:

1. **Disappearance of "Protrusion":** At the neutron density $\rho_{max} = 3 \cdot 10^{17}$ kg/m^3, even for one solar mass, the core already resides beneath the horizon (2.3 km vs 6 km). 2. **Field Expansion:** It is clearly visible that as mass increases, the process-freezing sphere (D_s) grows linearly, while the physical body (D_{core}) grows very slowly. 3. **The Gargantuan Void:** For supermassive objects (M87*), the horizon is over 16 million times larger than the physical object. This confirms that what we perceive is almost entirely "field atmosphere."

[

Structure of a Cold Star: The Core as a Field Generator] Structure of a Cold Star:
The Core as a Field Generator.

To avoid interpretational errors, one must distinguish the physical building block of the object from its inertial range:

- **Cold Core (D_{core}):** This is the "seed" of the Cold Star. An accumulation of matter with a density on the order of 10^{18} kg/m^3, forming the gravitational backbone of the object. This is where 100% of the real mass is located, triggering the dilation.

- **Freezing Sphere (D_s):** This is exclusively the range of the field generated by the Core. Within this region, an atom cannot vibrate ($T_m = 0$), which an external observer interprets as a "black hole."

- **Parasitic Mass:** Matter located in the space between D_{core} and D_s does not increase the freezing sphere linearly; it is merely subjected to the Core's field. The Core acts as the "fuse" that prevents parameters from tending toward infinity.

Conclusion: A black hole is not a "void with a horizon," but a **rigid core with a gargantuan dilation field**. The horizon is merely a "shadow" cast by a small, ultra-dense nucleus.

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21 Detailed Rotation Analysis of Sgr A* in the MRG-T Model

The following calculations demonstrate the evolution of rotational parameters from the neutron star stage to a supermassive Cold Core.

1. Boundary Conditions and Conservation Principle

We assume a model neutron star $M_1 = 3M_\odot$ with a radius $R_1 = 1.68$ km and a rotation frequency $f_1 = 500$ Hz. The conservation of angular momentum during mass growth to the level of Sgr A* ($M_2 = 4.1 \cdot 10^6 M_\odot$) is defined by the relation:

$$f_2 = f_1 \cdot \frac{M_1}{M_2} \cdot \left(\frac{R_1}{R_2}\right)^2 \tag{21}$$

2. Calculations for Sgr A*

- **Cold Core Radius (R_2):** 98.6 km (at $\rho = 2 \cdot 10^{18}$ kg/m^3).
- **Rotation Frequency (f_2):**

$$f_2 \approx 500 \cdot 7.35 \cdot 10^{-7} \cdot 0.000289 \approx 1.06 \cdot 10^{-7} \text{ Hz} \tag{22}$$

- **Tangential Surface Velocity (V_{eq}):**

$$V_{eq} = 2\pi \cdot f_2 \cdot R_2 \approx 0.065 \text{ m/s} \tag{23}$$

3. Physical Conclusion

The calculated velocity of **6.5 cm/s** forms the bedrock of the Cold Star paradigm. In contrast to the relativistic models of GR (where $V \approx c$), the MRG-T model predicts that galactic centers are objects of extreme inertial stability. The 10^9 order of magnitude difference between the predictions of both theories constitutes a measurable "scientific problem" that must be resolved through interferometric observations.

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Variability of Angular Momentum and Real Rotational Velocities

It must be emphasized that the presented values (order of 10^{-2} m/s) are model-based. The actual rotational velocity of a Cold Star is an

individual parameter, dependent on the object's evolutionary history.

- **Starting Parameter Variance:** Observed neutron stars range from fractions of a second to several seconds per revolution.
- **Inertial Conclusion:** Regardless of the initial conditions, the MRG-T model precludes the existence of relativistic surface velocities ($V \approx c$).
- **Stability of the Foundation:** Even with an extremely rapid rotation of the progenitor star, the gargantuan increase in the moment of inertia (scaling to millions of solar masses) "extinguishes" the tangential surface velocity to values that are completely benign from the perspective of classical physics.

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22 Summary and Empirical Verification

The credibility of the MRG-T model is not based on theoretical speculation, but on its ability to precisely describe reality across vastly different scales. The following table contrasts the model's predictions, based on the inertial constant $\alpha = 0.85$, with observational data.

Final Conclusion

The end of the era of singularities is a mathematical fact. The constancy of the $\alpha = 0.85$ coefficient and the demonstration that the photon lacks massive momentum closes the chapter of physics based on black holes as "holes" in spacetime.

The Inertial Cold Star is a real, measurable object with a temperature of 0 K. Instead of consuming the Universe, it serves as its hardest foundation and most efficient energy transmitter. Anyone who chooses Einstein's infinities over the measurable stability of the MRG-T model chooses dogma over science.

Numbers do not lie. It is time to return to reality.

Object / Scale	**MRG-T Prediction ($\alpha = 0.85$)**	**Observations / GR**
The Moon (ΔT_m)	Process pulse difference: **7 μs / day**	To be verified (Artemis)
Mercury (Precession)	**43.0"** / century (logarithmic correction)	43.1" / century
M87* (Jet Power)	4×10^{36} **W** (0 K absorption rejection)	$10^{35} - 10^{37}$ W
Neutron Star	Cold core diameter: **1.4 km** ($\kappa \approx 9.6$)	Pulsar stability

23 Demythologizing Dilation: Biological Limit and Universal Time

The MRG-T model assumes the existence of **Pure Time (T_C)** as the absolute reference frame, identified with the cosmic vacuum (minimum inertial resistance). Every other frame is slowed down relative to Pure Time by the coefficient κ.

The Millennium Experiment Calculation

To demonstrate the impossibility of "escaping into the future," we analyze the theoretical case of an observer remaining in a field of maximum inertial saturation ($\kappa_{limit} \approx 9.64$) for a period of 1,000 years as measured on Earth.

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Conversion to Pure Time (Absolute Standard)

Earth, as a system with a coefficient of $\kappa_Z \approx 1.45$, slows down physical processes relative to the vacuum. Therefore, 1,000 Earth years correspond to the following interval in Pure Time:

$$T_C = 1,000 \text{ years}_Z \times 1.45 = 1,450 \text{ years}_{Pure} \tag{24}$$

The Observer's Biological Age

During this same universal interval, the biological processes of an observer near a Cold Star occur at a rate reduced by κ_{limit}:

$$T_{bio} = \frac{1,450 \text{ years}_{Pure}}{9.64} \approx 150.4 \text{ years} \tag{25}$$

23.1 Comparative Table of Time Passage

Reference Frame	**Time Passage (Years)**	**Human Biological Status**
Pure Time (Vacuum)	1,450	Physical end (natural death)
Earth (Calendar)	1,000	Physical end (natural death)
Cold Star (Limit)	**150.4**	**Life limit exceeded (return impossible)**

Table 1: Comparison of organic matter aging in different inertial fields assuming 1,000 Earth years.

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Conclusion

The maximum slowdown of biological processes relative to Earth is approximately 6.6-fold (κ_{limit}/κ_Z). Since no human organism can

survive for 150 years in an active state, no one can "return" from the vicinity of a Cold Star after 1,000 Earth years. Nature, through the finite value of κ, prevents time travel by reducing dilation to the role of a physical process brake rather than a gateway to the future.

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The photon has no momentum. pdf

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